Oceanus (Zhuoyang) Zhang

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SKILLS

Programming Languages & Tools: R, Python, SQL, Bash, Git, AWS, Plotly, R Shiny, LaTeX
Analytics & Modeling: Machine Learning, GLM, logistic regression, mixed-effects models, PCA, GSEA, limma/voom,
Data & Engineering: ETL pipelines, graph databases (Neptune), REST APIs, data wrangling, reproducibility
Domains: Omics / proteomics, clinical outcomes, health data, customer behavior analytics

EDUCATION

University of California, Merced *(Merced CA)* — *July 2026 - Present*
- *Doctor of Philosophy in Computational Biology*

Emory University, Rollins School of Public Health *(Atlanta GA)* — *August 2023 - May 2025*
- *Master of Public Health in Biostatistics & Bioinformatics | GPA: 3.75*
- *Data Sciences Certificate*

University of California, Irvine *(Irvine CA)* — *August 2020 - June 2023*
- *Bachelor of Science in Biological Sciences **(Honors)** | GPA: 3.60*

WORK EXPERIENCE

Bioinformatics Intern II, *Alamar Biosciences Department of Bioinformatics (Fremont, CA)* — *May 2025 - Dec 2025*
Under the supervision of Joanne Beer, PhD
- **Diagnosed plate-effect drivers** by correlating signal-quality metrics, environmental conditions, and sample-stability data, helping improve **assay reproducibility** across runs
- Engineered and deployed an **R Shiny + Plotly dashboard** for **plate randomization and workflow tracking** on **AWS**, enabling standardized experiment setup and cross-team visibility
- Built an **FDR-aware power & sample-size Shiny tool** that **automated study-design decisions** for large-scale omics experiments and reduced manual analysis time

Data Science Intern, *Zymo Research Corp Department of Integrated Omics (Tustin, CA)* — *May 2024 - May 2025*
Under the supervision of Kyle McClary, PhD
- **Designed and populated** an **AWS Neptune knowledge-graph database** linking researchers, diseases, and microbiome entities, enabling relationship-driven data exploration
- **Developed a Python ETL and analytics pipeline** for personalized diet profiling using HEI food-group mapping and energy-normalized intake metrics
- **Built and validated logistic-regression purchase-prediction models** (~75% accuracy) and integrated API-based mailing automation, contributing to $32K in campaign revenue within one week

Machine Learning Intern, *UC Irvine Department of Computer Sciences (Irvine, CA)* — *March 2024 - Sep 2024*
Under the supervision of Jing Zhang, PhD
- Designed a modular **Machine Learning pipeline** for data preprocessing, model training, missing-region completion, and evaluation across **spatial-omics datasets**
- Implemented and benchmarked multiple completion approaches (**random, KNN, GAN, latent-space GAN**) to establish model baselines
- Evaluated model performance using Chamfer Distance, Earth Mover's Distance, MSE, MAE, and Cosine Similarity
- Produced reproducible, publication-ready results and tables in LaTeX to support manuscript submission

RESEARCH EXPERIENCE

Clinical Research Analyst, *Emory University School of Medicine (Atlanta, GA)* — *July 2024 - July 2025*
Under the supervision of Amber Kulshreshtha, MD, PhD

- Applied **logistic regression, GLMs, and mixed-effects** models to analyze patient outcomes in **NIH-funded studies** on antibiotic prescribing and diabetes readmissions
- Automated R workflows to generate **publication-ready tables and visualizations,** accelerating manuscript development and stakeholder reporting

Bioinformatics Research Assistant - *Emory University School of Public Health (Atlanta, GA)February 2024 - May 2025*
Under the supervision of Xiangqin Cui, PhD and Christopher Willey, MD, PhD (Capstone Project)

- Processed and harmonized gene-expression datasets (25K+ genes) for GBM xenolines under hypoxia vs normoxia
- Ran differential-expression and GSEA pipelines to identify pathway-level responses
- Built an interactive Shiny app for clinicians to query enriched pathways by xenoline and condition

Graduate Researcher, *Emory University School of Medicine (Atlanta, GA)* *May 2023 - April 2024*
Under the supervision of Aliza Wingo & Thomas Wingo, MD, PhD

- Developed a reproducible voom-limma analysis pipeline for miRNA expression across Alzheimer's, Depression, and MCI cohorts
- Performed outlier detection, GSEA, and visualization and drafted analytic findings for Applied Practical Experience review

Undergraduate Researcher, *UC Irvine School of Biological Sciences (Irvine, CA)* *March 2022 - June 2023*
Under the supervision of Monica Daley, PhD

- Led field-based data collection and applied PCA and mixed-effects modeling to study behavioral and morphological traits in kangaroo rats
- Presented findings at SICB 2023

PUBLICATION

Article

- Zhang, Z. et al. *DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion*, *IEEE International Conference on Image Processing,* Aug, 2025 (Co-author, Accepted)
- Zhang, Z. et al. *Association of Neighborhood Location and 30-Day Readmissions in Patients with Diabetes*, *Journal of Hospital Medicine,* July 2025 (Co-author, Accepted)

Conference Abstract

- Zhang, Z. et al. *Telehealth-based Stress Reduction for Mild Cognitive Impairment Patients, Alzheimer's Association International Conference, AAIC 2025* (Co-author, Accepted)
- Zhang, Z. et al. *Telemedicine Versus In-Person Primary Care Visitors for Upper Respiratory Infections: Comparison of Antibiotic Prescribing, North American Primary Care Research Group, NAPCRG 2025* (Co-author, Accepted)

CLASS PROJECTS

Breast Cancer Wisconsin Dataset - Applied machine learning techniques including logistic regression, and optimized models through cross-validation and hyperparameter tuning. Evaluated model performance with ROC curves and feature importance plots

Anemia Prediction in Pregnant Women Study - Applied machine learning models including logistic regression, and conducted robust model evaluations using ROC curves and AUC metrics, to analyze the influence of C-reactive protein and micronutrient levels on anemia in pregnant women

Statistical Analysis of the Prospective Evaluation of Radial Keratotomy (PERK) Study - Conducted statistical analysis using SAS on radial keratotomy (RK) data in treating myopia, including T-tests, ANOVA, Chi-square, and regression analysis, to explore the impact of RK across different patient groups.